Chunfan **Yao**

Undergraduate Researcher · AI Agents and Verification

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Summary

HKUST(GZ) AI undergraduate and Fall 2026 visiting student at UC Berkeley. I build evaluation environments and executable checks for coding and research agents. I am interested in applying these methods to agent security and program analysis.

Education

The Hong Kong University of Science and Technology (Guangzhou) — *Guangzhou, China*

B.Eng. in Artificial Intelligence — *Sep. 2024 – Present*

University of California, Berkeley — *Berkeley, CA*

Visiting Undergraduate (Exchange) — *Aug. 2026 – Dec. 2026*

Research Experience

Unit-Weight Gadgets for Rydberg-Atom Arrays — Prof. Jinguo Liu's Group, HKUST(GZ) — *Guangzhou, China*

Undergraduate Research Assistant (Co-First Author) — *2025 – Present*

- Proposed a tensor-preserving vertex-split construction to reduce pin degree in unit-weight gadgets for triangular Rydberg lattices.
- Implemented reduced-α tensor checks, lattice-animal enumeration and port validation in GadgetSearch.jl. A manuscript is in preparation.

LLM Combinatorial Reasoning Benchmark — Prof. Jinguo Liu's Group, HKUST(GZ) — *Guangzhou, China*

Research Contributor — *2025 – 2026*

- Built early SWE-style and mini-SWE-agent runners for problem-reductions-benchmark, including a common agent interface, concurrent execution and trajectory archiving.
- Developed certificate-based checks with the `pred` verifier for subtle logical errors in combinatorial reductions; defined token- and cost-normalized evaluation metrics.

AutoResearch for Reduction-Rule Discovery — HKUST(GZ) — *Guangzhou, China*

Research Workflow Design — *2026 – Present*

- Designed the problem-finding component of a verify-first workflow: explicit polynomial-time contracts, retained counterexamples and independent checks. The ongoing project has generated 33 candidate rules; it is not yet fully public.

Selected Project

Reduction Impact Atlas — *Open Source*

Independent Research Tool — *2026*

- Built a reproducible ranking of 76 open reduction questions using frozen graph data and explicit bridge evidence. Scores estimate conditional reachability; unverified mappings remain labeled provisional. Source code.

Technical Skills

Languages	Python, Julia, Git, Linux/CLI, LaTeX
Agent/Eval	Agent evaluation harnesses, executable verifiers, benchmark design, trajectory analysis
Methods	Graph algorithms, combinatorial optimization, PyTorch, experiment automation